

Collin Bolles

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Education

Boston University

M.S. Robotics and Autonomous Systems

Rochester Institute of Technology

B.S. Computer Science

Employment

Software and Application Innovation Lab

Senior Fullstack Software Engineer

Nov. 2025 to Present

Fullstack Software Engineer

May 2022 to Nov. 2025

- Produced a data labeling platform resulting in largest of it's kind sign language datasets
- Handled migration of 20+ projects from RedHat OpenShift to AWS avoiding client application downtime
- Deployed custom LLM used in a global disease outbreak detection system
- Provided technical expertise supporting a ROS2 ecosystem along with integrating legacy ROS1 platforms
- Produced GitHub action CI/CD pipelines for rapid software deployment turnaround
- Tools and Technology: AWS, Kubernetes, OpenShift, Typescript, React, NestJS, MongoDB, PostgreSQL, ROS1, ROS2

D3 Engineering · Embedded Engineering Technician

Jan. 2021 to Dec. 2021

- Resolved hardware compatibility resulting in higher FPS for client robotics application
- Developed Linux hardware drivers to bring computer vision to real time robotics systems
- Customized NVIDIA/Linux Kernel to enable camera frame synchronization
- Debugged newly developed hardware systems
- Tools and Technology: C, Python, Linux Kernel, Git, Gerrit, Logic Analyzer

Herrick Technology Labs · Software Intern

Sep. 2020 to Dec. 2020

- Worked on an inter-disciplinary team tasked with producing a multi-faceted sensing solution for US military avionics
- Wrote libraries to communicate with HTL radio solutions over a range of protocols including TCP over Ethernet and UART
- Tools and Technology: C++, Python, Git, Cuda

Blue Spiral · Software Developer

June 2020 to Aug. 2020

- Wrote iOS application for on-the-spot employee performance reviews using SwiftUI
- Modernized OCR system using Azure OCR API
- Leveraged satellite imagery and computer vision techniques for monitoring vegetation health
- Tools and Technology: C#, Swift, Python

Ball Bowler · Software Developer

Jan. 2019 to Nov. 2019

- Brought a proof-of-concept UI to a production ready system
- Designed and developed a UI for a miniature bowling lane
- Captured state of bowling pins using computer vision
- Tools and Technology: Java, JavaFX

Blue Spiral · Software Intern

Aug. 2017 to Aug. 2018

- Developed object detection training pipeline built on top of Tensorflow
- Applied object detection pipeline for the detection of unwanted ducks
- Incorporated object detection pipeline with the Microsoft Hololens
- Tools and Technology: C#, Python, C++, Tensorflow

Valogix · Software Intern

July 2016 to Nov. 2018

- Created custom monitoring solution allowing for insight into 100+ deployed applications
- Resolved bugs and incorporated features in the existing Valogix code base
- Tools and Technology: Java, Spring, Spring Boot, PostgreSQL, Groovy

Activities

Electric Vehicle Team

Firmware Lead

May 2020 to May 2022

Firmware Member

Dec. 2018 to May 2020

- Firmware development for custom built electric motorcycles
- Created firmware for a high-demand battery management system
- Produced a highly adaptable software interface layer for the STM32 platform
- Developed drivers across communication protocols (CAN, UART, I2C, PWM)
- Unified the inter-board communication using CANopen